

Runtime-Constrained Residual Execution for Sequential Dual-Sided Grasping

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Abstract—A dual-sided dexterous hand can retain one object while acquiring another, but shared wrist motion couples the grasps. We formulate second-stage execution as realized-state-conditioned residual control with separated execution authority. An upstream planner fixes a nominally constraint-checked suffix; Soft Actor-Critic (SAC) adjusts only retention-side references and path progress; and deterministic modules constrain or reject commands. This separation keeps nominal task intent outside the learner, providing a candidate execution layer for future shared autonomy. Across 40 simulated episodes per package spanning eight ordered object pairs, SAC with domain randomization, SAC, and the deterministic package satisfy the composite success criterion in 35 (87.5%), 33 (82.5%), and 28 (70.0%) trials. Overlapping Wilson intervals support preliminary package-level feasibility rather than statistically significant superiority. Human-in-the-loop and hardware evaluation remain future work.

I. INTRODUCTION

Sequential multi-object grasping assigns different fingers to successive objects [1], while reversible hands can grasp from opposing sides [2]. We study *sequential dual-sided grasping*: retention side R holds O_R while side A acquires O_A (Fig. 1). The sides have disjoint actuators but share the wrist and arm trajectory. First-grasp errors can therefore turn nominally feasible second-stage motion into retained-object drift.

Sequential grasp generators may predict subsequent poses [3]; residual RL can correct nominal controllers [4] and support multi-fingered grasping [5]; and deterministic filters can restrict learned actions [6]. We allocate authority differently: g_A^* and τ_A^* remain fixed, the learner changes only retention references and path progress, and deterministic modules project or reject commands. Keeping nominal task intent outside the learner makes this execution-layer interface compatible with future shared autonomy. We contribute a realized-state-conditioned formulation, an authority-separated interface, and a randomized-simulation characterization of three complete execution packages.

II. REALIZED-STATE-CONDITIONED EXECUTION

A. Task and Residual Policy

After the first grasp, the realized boundary state B_0 records the robot state, retained hand–object transform $\hat{T}_{R,0}$, tracking quality, and object poses. An upstream planner selects g_A^* and returns τ_A^* after nominal collision, kinematic, and joint checks. Planning lies outside our contribution; these nominal checks do not cover execution under model mismatch.

During execution, a model-based RGB-D registration tracker estimates $\hat{T}_{R,t}$ without semantic masks or simulator pose labels. The actor, projector, and runtime monitor receive no simulator contact states. SAC observes RGB-D, proprioception, path progress, tracking quality, and $\hat{T}_{R,0}^{-1}\hat{T}_{R,t}$. Its action is $a_t = (\Delta q_{R,t}, \alpha_t)$, where $|\Delta q_{R,t}^{(j)}| \leq 5^\circ$ modifies only retention joints and $\alpha_t \in [0.25, 1]$ scales progress along the fixed suffix.

We train SAC [7] in Isaac Lab [8]. The reward balances retention, path progress, terminal dual hold, drift, smoothness, slowdown, projection, and object loss. Episode-constant domain randomization [9] uniformly scales mass by $[0.9, 1.1]$, dimensions by $[0.98, 1.02]$, and friction by $[0.8, 1.2]$. It scales actuator/controller parameters by $[0.9, 1.1]$ and offsets each center-of-mass coordinate within ± 2 mm. Five training seeds each use 10^7 transitions. Checkpoints are selected on a fixed randomized validation set disjoint from training, and we report the seed with median validation performance.

B. Deterministic Authority Boundary

At 20 Hz, the proposed action is projected by

$$\tilde{a}_t = \arg \min_{a \in \mathcal{C}(o_t)} \frac{1}{2} \|W(a - a_t)\|_2^2, \quad (1)$$

where $W = \text{diag}(1, \dots, 1, 0.5)$. The set $\mathcal{C}(o_t)$ imposes a 3° joint-limit margin, retention-side velocity, acceleration, and jerk limits of 0.35 rad/s, 2.0 rad/s², and 20 rad/s³, and $|\alpha_t - \alpha_{t-1}| \leq 0.10$. Infeasibility or a 2 ms timeout triggers a runtime hold.

The runtime monitor rejects arm-joint deviations above 5° , end-effector deviations above 15 mm or 5° , retained-object drift above 15 mm or 15° , invalid tracking, or observations older than 100 ms. Rejection sets $\alpha = 0$, holds the last accepted retention command for 0.5 s, and terminates the trial. These mechanisms provide command-level constraints and threshold rejection, not formal state-level guarantees.

III. PRELIMINARY SIMULATION RESULTS

We compared three selected execution packages. The deterministic package sets $\Delta q_R = 0$ and uses a deterministic progress schedule; SAC and SAC+DR learn both action channels. All packages share B_0 , g_A^* , τ_A^* , observations, projector, monitor, and command limits. Eight ordered object pairs and five paired test seeds, disjoint from training and validation, give 40 episodes per package. The composite

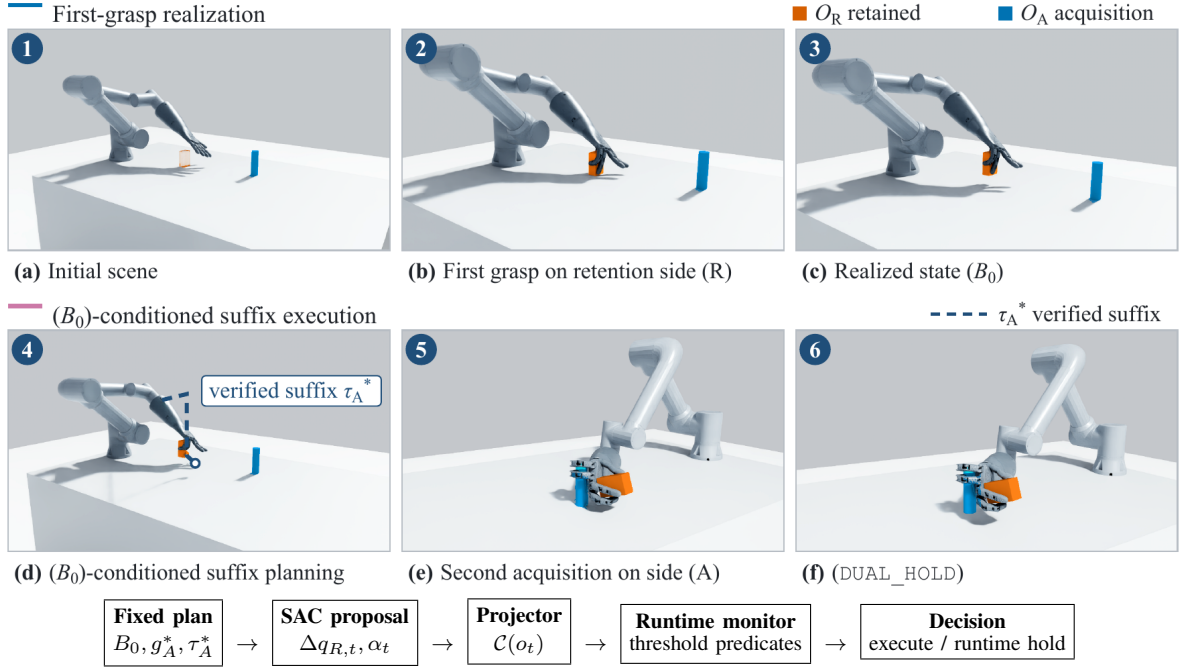


Fig. 1. Sequential dual-sided grasping and execution authority. The realized state B_0 anchors a fixed suffix; SAC adjusts retention references and progress; deterministic modules project or reject commands.

success criterion requires suffix completion, a 3 s terminal DUAL_HOLD, drift within 15 mm and 15°, and monitor acceptance.

SAC+DR, SAC, and the deterministic package met this criterion in 35/40 (87.5%), 33/40 (82.5%), and 28/40 (70.0%) trials. Their episode-level Wilson 95% intervals were 73.9–94.5%, 68.1–91.3%, and 54.6–81.9%, respectively, and overlapped. Runtime monitoring terminated 2, 3, and 6 trials, while median maximum translation drift among successful trials was 8.5, 9, and 11 mm. Across paired test seeds, projection-intervention rates ranged from 4–10%, 5–12%, and 0–3%. These descriptive package-level results do not isolate the effects of residual control, speed adaptation, randomization, projection, or monitoring.

IV. LIMITATIONS AND OUTLOOK

The evidence covers one hand, one simulated task family, and one selected policy per SAC variant, assuming valid B_0 and τ_A^* . It therefore supports preliminary package-level feasibility within the tested setting. We do not evaluate operator input, workload, situational awareness, or intervention demand. Future work will integrate externally supplied task intent, evaluate human-in-the-loop authority allocation, formalize execution contracts, and extend the study to hardware.

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